

Homework 1: Exploratory Data Analysis

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Dataset choice

For this assignment, we will be using the Chemical Manufacturing Process dataset. This dataset was taken from Applied Predictive Models (APM) Chapter 1.4.

Data set fields description

Outcomes:

- **Yield** (continuous) = Final product yield. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	35.25	38.75	39.97	40.18	41.48	46.34

Predictors:

- **BiologicalMaterial01** (Continuous) = Measurement one of the biological starting material. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	4.580	5.978	6.305	6.411	6.870	8.810

- **BiologicalMaterial02** (Continuous) = Measurement two of the biological starting material. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	46.87	52.68	55.09	55.69	58.74	64.75

- **BiologicalMaterial03** (Continuous) = Measurement three of the biological starting material. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	56.97	64.98	67.22	67.70	70.43	78.25

- **BiologicalMaterial04** (Continuous) = Measurement four of the biological starting material. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	9.38	11.24	12.10	12.35	13.22	23.09

- **BiologicalMaterial07** (Categorical) = Measurement seven of the biological starting material, taking the values:

```
## [1] 100.00 100.83
```

- **ManufacturingProcess01** (Continuous) = Measurement one of the manufacturing process. A summary of this column is:

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.      NA's
##      0.00   10.80   11.40   11.21   12.15   14.10         1
```

- **ManufacturingProcess02** (Categorical) = Measurement two of the manufacturing process. Examples of some values include:

```
## [1]    NA    0.0 19.7 19.9 19.3 19.5
```

There are a total of

```
## [1] 28
```

values.

- **ManufacturingProcess03** (Continuous) = Measurement three of the manufacturing process. A summary of this column is:

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.      NA's
##      1.47   1.53   1.54   1.54   1.55   1.60        15
```

- **ManufacturingProcess04** (Continuous) = Measurement four of the manufacturing process. A summary of this column is:

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.      NA's
##      911.0  928.0  934.0  931.9  936.0  946.0         1
```

- **ManufacturingProcess12** (Categorical) = Measurement twelve of the manufacturing process, taking the values:

```
## [1]    NA    0 4549
```

CoNVO statement

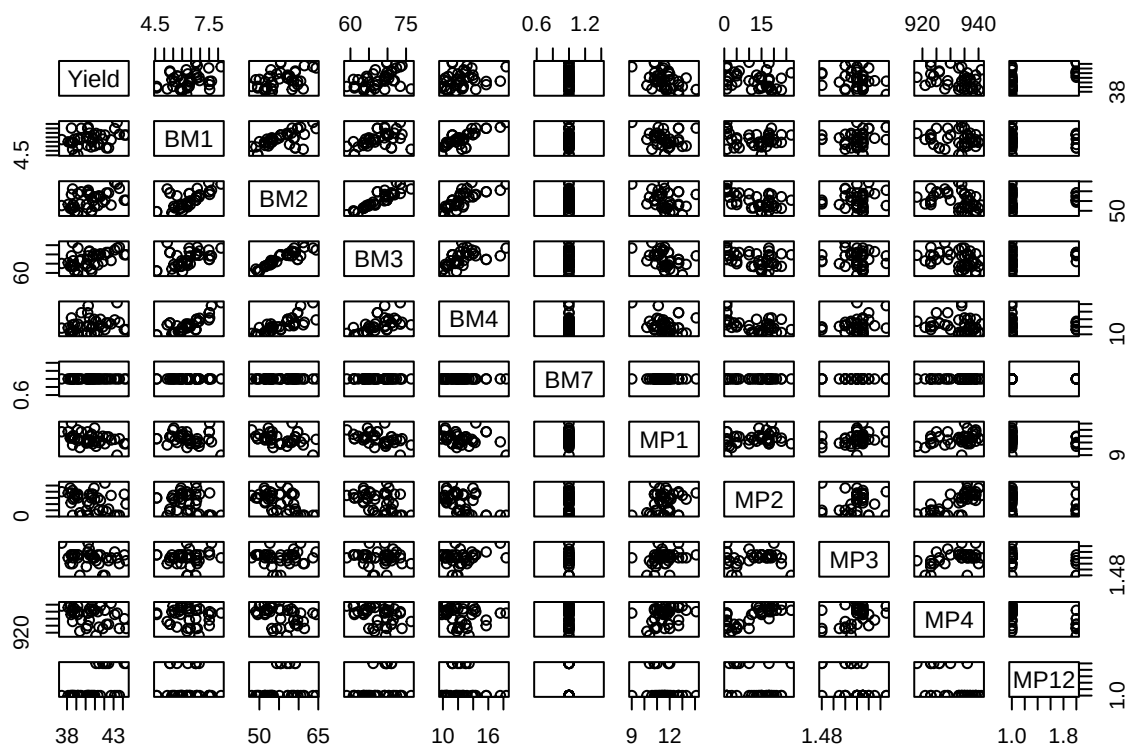
Pfizer is a pharmaceutical manufacturing company. Maximizing yield in manufacturing vaccines is important. The industrial engineering team at Pfizer is trying to understand what stages of their manufacturing process impact yields the most. They will use this information to make strategic decisions about what stages of the process need upgraded equipment to improve accuracy.

Manufacturing vaccines is a very complicated process. There are many sequential operations, many of which impact the yield of a batch. Pfizer wants to determine which factors make the most difference. The difficulty is that there are complex interactions between factors, and manufacturing processes differ by the material properties.

The project will use historical data on process yields (Chemical Manufacturing Process dataset) to explore the effects of each factor on yield. These relationships will be visualized through several plots to find potential relationships. If a relationship is found, it will be reported to the IE team who will determine how to improve that process or material property for a larger yield.

If the head of the Pfizer IE team is happy with initial results, a deeper analysis will be conducted to determine which interactions between material properties and manufacturing processes have the largest impact on yields. This analysis will be delivered to them every quarter to determine growth opportunities when it comes to yields.

Pairs plot

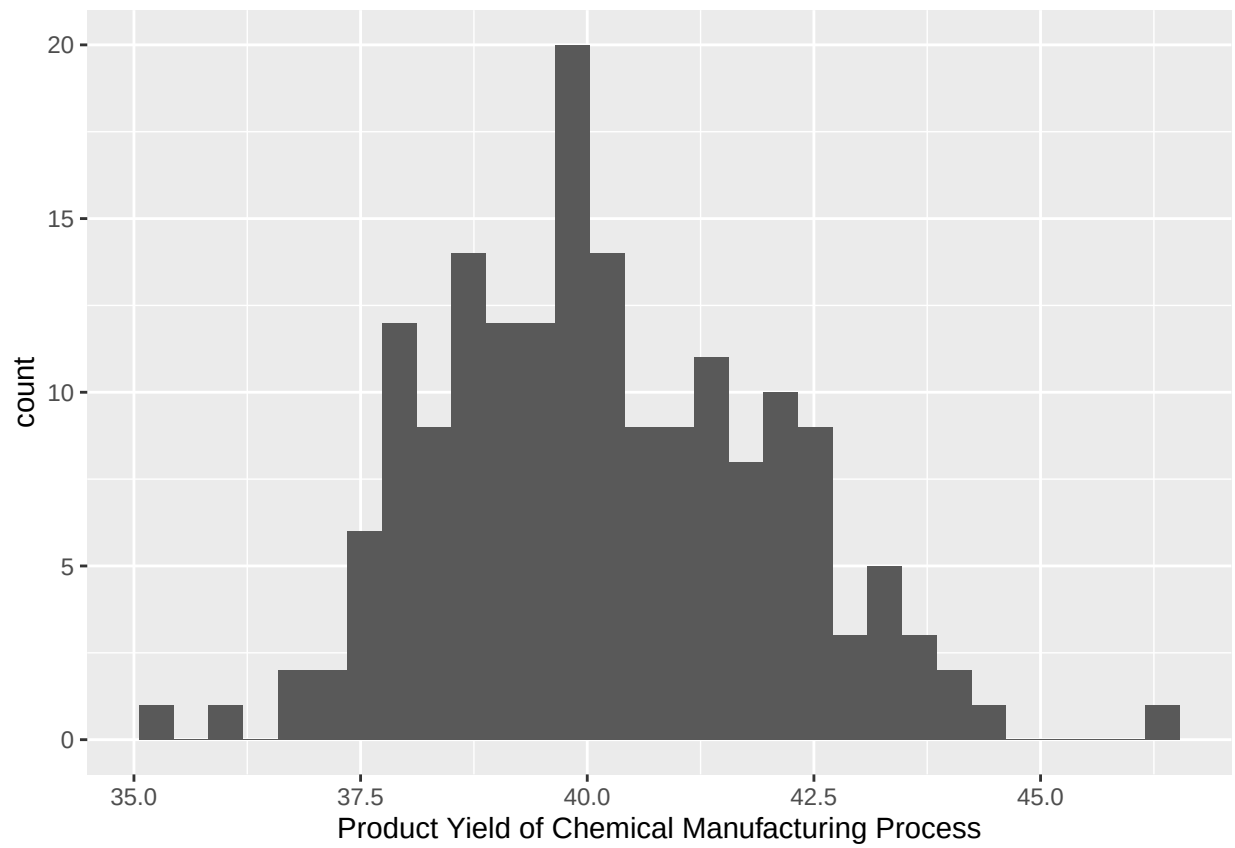


Identify an outcome and three predictor variables

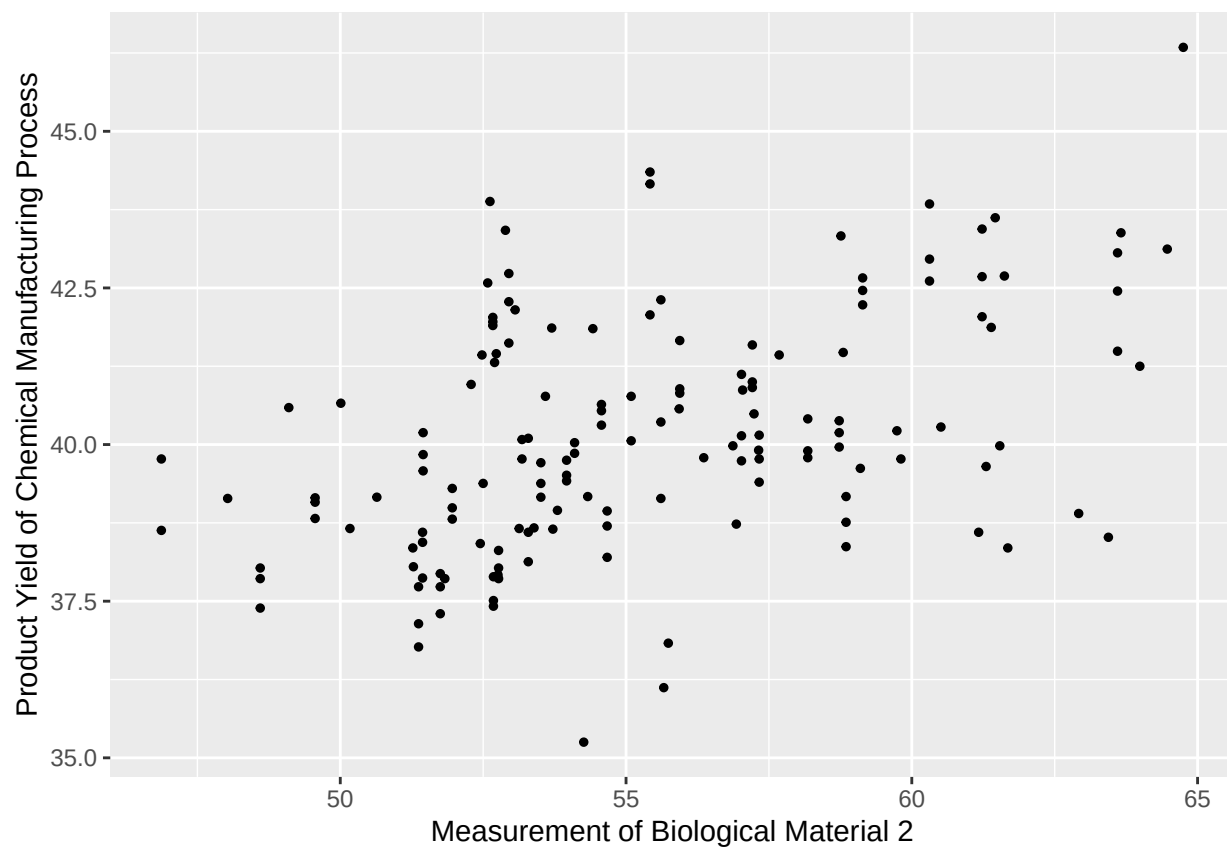
The outcome variable is yield.

We will choose BiologicalMaterial02, ManufacturingProcess02, and ManufacturingProcess12 as they appear uncorrelated with each other and are highly correlated with the outcome.

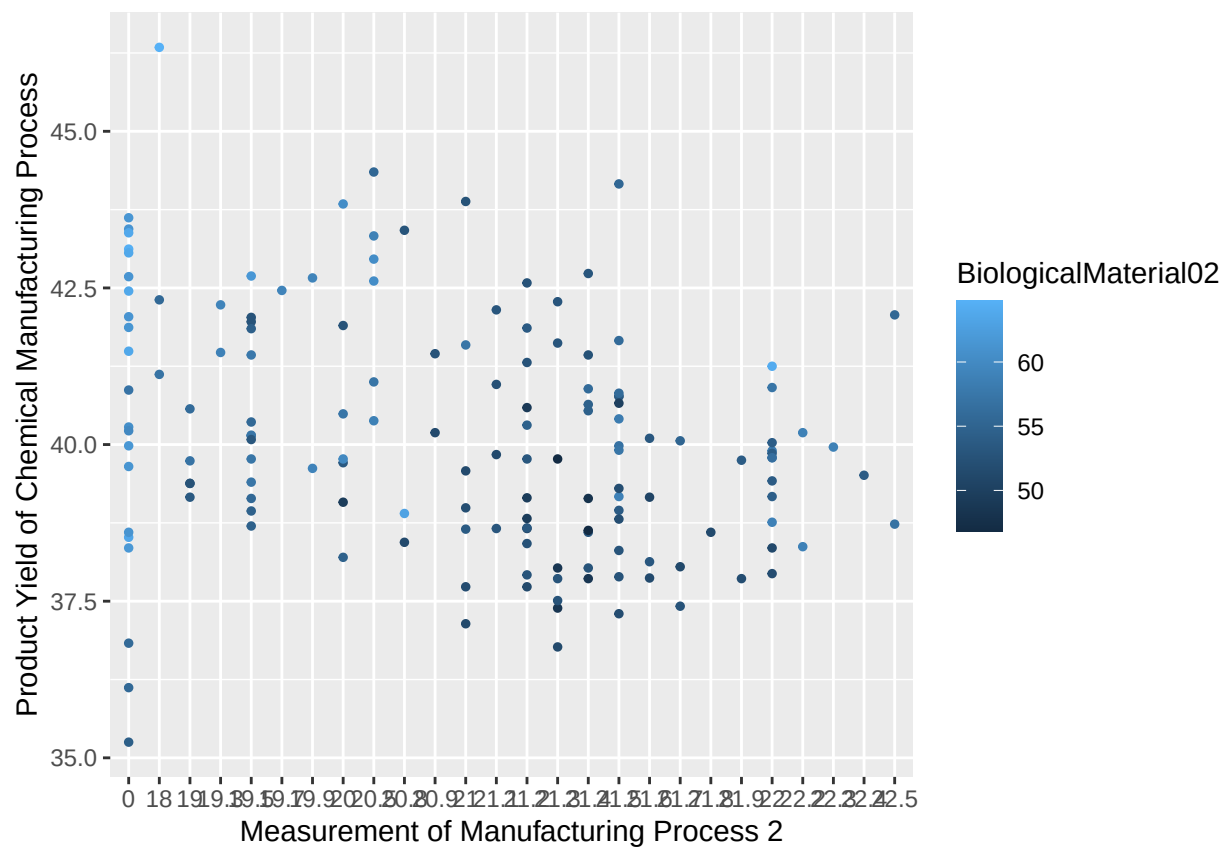
Explore the variables by applying four plot types



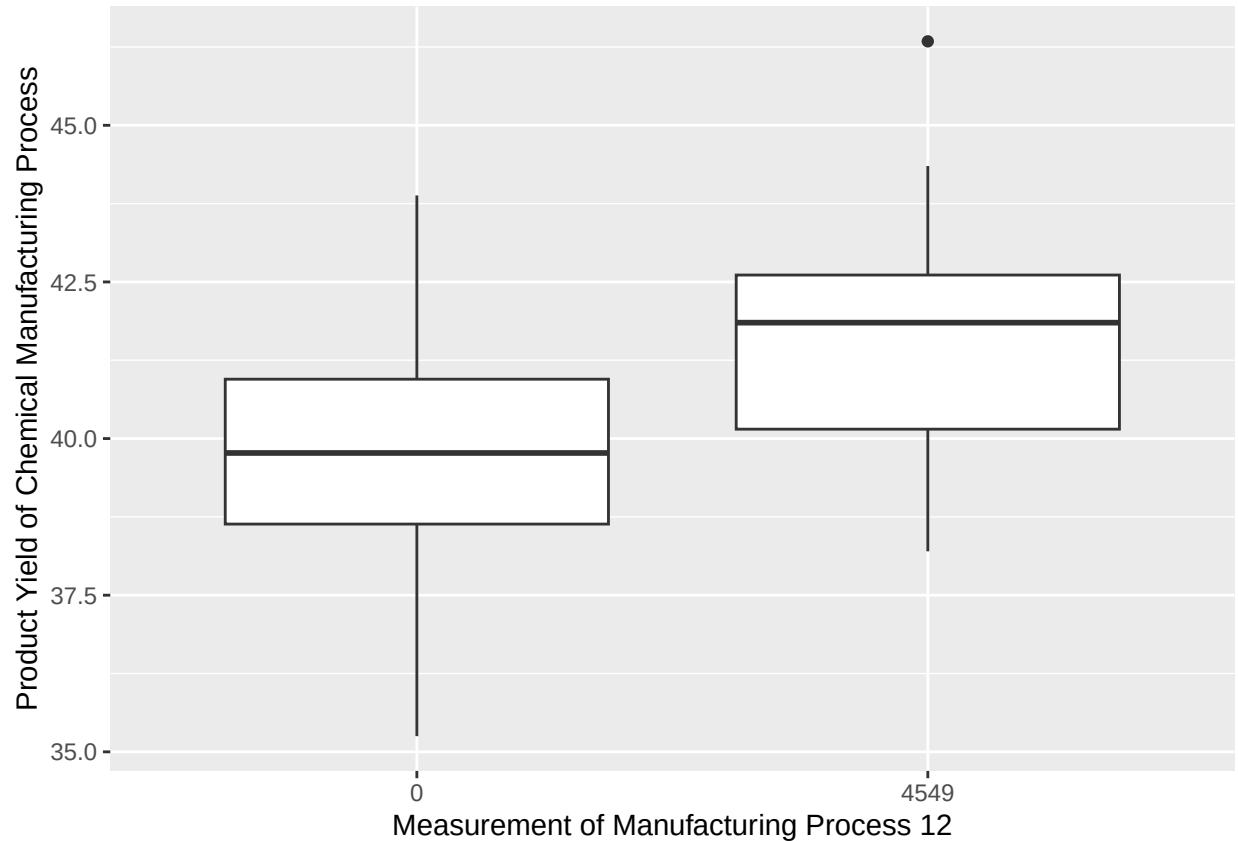
The histogram above is used to visualize the distribution of Yield values. Most fall between 37.5 and 42.5.



Generally, there is a positive correlation between Biological Material 2 and Yield. When the measurement for Biological Material 2 is higher, Yield tends to be higher.



This graph shows the interaction between Biological Material 2 and Manufacturing Process 2 as it related to the Yield. The majority of measurements for process 2 are between 17.5 and 22.5, with about 10 at 0. The points that are non-zero tend to have similar yields to the zero values.



When Manufacturing Process 12 is measured at 4549, the mean and right tail are larger than when it is measured at 0.

Causal relationships

We see a positive relationship between biological material 2 and yield. There is likely some causality there, though it's unclear as there's no information about what biological material 2 actual measures.

There doesn't seem to be much of a relationship between manufacturing process 2 and yield. Measurements of zero and all other measurements didn't seem to make a difference in yield. There may be a slight negative relationship in the non-zero group of measurements in manufacturing process 2. Similar to above, there is no information about what manufacturing process 2 is measuring, which makes it difficult to determine causality.

There seems to be a relationship between manufacturing process 12 and yield. When manufacturing process 12 was measured at 0, the mean yield was about 2 less than when it's measured at 4549. Again, it's difficult to determine causality since it's unclear what the variables are measuring.